

Lecture 3

A Brief Review of Linear Models

3.1

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See Fahrmeir L., Kneib T., Lang S. (2007) Regression. Modelle, Methoden und Anwendungen, Springer. (2013) Regression. Models, Methods and Applications, Springer.	

3.2

1 The Linear Classical Regression Model

Data

$$(y_i, x_{i1}, \dots, x_{ip}), \quad i = 1, \dots, n,$$

with y metric variable and $x_1, \dots, x_p$ metric predictors or categorical regressors recoded as binary variables.

3.3

Model

The model for the i th subject is

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n,$$

The errors $\varepsilon_1, \dots, \varepsilon_n$ are assumed to be uncorrelated¹

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ when } i \neq j$$

¹ Sometimes the stronger assumption of independent and identically distributed (i.i.d.) errors is done.

with

$$\begin{aligned} E(\varepsilon_i) &= 0, \quad i = 1, \dots, n, \\ \text{or } E(\varepsilon_i | x_{i1}, \dots, x_{ip}) &= 0, \quad i = 1, \dots, n, \text{ with stochastic regressors,} \\ \text{Var}(\varepsilon_i) &= \sigma^2, \quad i = 1, \dots, n. \end{aligned}$$

The estimated linear function

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$$\hat{f}(x_1, \dots, x_p) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

can be seen as the estimate $E(y | x_1, \dots, x_p)$ of the expected value of y given the values $x_1, \dots, x_p$ of the explanatory variables, and can be used to predict y .

Predictions will be denoted with $\hat{y}$.

3.5

In matrix form we have

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ are $(n \times 1)$ vectors, $\mathbf{X}$ is a $(n \times (p + 1))$ matrix and $\boldsymbol{\beta}$ is a $((p + 1) \times 1)$ vector

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

3.6

The Linear Classical Regression Model

The model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

is a classical regression model when the following conditions hold

1. $E(\boldsymbol{\varepsilon}) = \mathbf{0}$,
or $E(\boldsymbol{\varepsilon} | \mathbf{X}) = \mathbf{0}$ when regressors are stochastic,
2. $\text{Var}(\boldsymbol{\varepsilon}) = E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') = \sigma^2 \mathbf{I}_n$.
3. $\mathbf{X}$ has full column rank $r(\mathbf{X}) = p + 1$.
4. $E(\mathbf{X}'\boldsymbol{\varepsilon}) = 0$.

If the assumption

$$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

$$\text{or } \boldsymbol{\varepsilon} | \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n) \text{ when regressors are stochastic}$$

holds, we are in presence of a Linear Gaussian Classical Regression Model.

In this case we have i.i.d. errors.

3.7

2 Estimation

Estimation

With Ordinary Least Squares (OLS), as well as with maximum likelihood, $\boldsymbol{\beta}$ is estimated as

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}.$$

Predicted values of $\mathbf{y}$ corresponding to the training data set $\mathbf{X}$ can be obtained as

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$$

with $\mathbf{H}$ the so-called *hat matrix*

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

The *OLS estimator* of the variance σ^2 common to the errors ε_i is

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}},$$

where

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

3.8

Properties of OLS Estimator

- $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$,
- $Var(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$,
- $E(\hat{\sigma}^2) = \sigma^2$.

$Var(\hat{\boldsymbol{\beta}})$ can be estimated as

$$\widehat{Var}(\hat{\boldsymbol{\beta}}) = \frac{1}{n-p-1} \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} (\mathbf{X}'\mathbf{X})^{-1}.$$

3.9

Estimator Properties for the Linear Gaussian Classical Regression Model

If the normality assumption holds we have

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1})$$

and

$$\frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' (\mathbf{X}'\mathbf{X}) (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\sigma^2} \sim \chi_{p+1}^2.$$

The preceding properties of a Linear *Gaussian* Classical Regression Model hold irrespective of the sample size.

3.10

When the Gaussian assumption is not satisfied, we can have recourse to the following asymptotic properties in order to evaluate estimators.

Asymptotic Properties of Estimators for the Linear Classical Regression Model

If

$$\lim_{n \rightarrow \infty} \frac{1}{n} (\mathbf{X}'_n \mathbf{X}_n) = \mathbf{V},$$

with $\mathbf{V}$ positive definite matrix, we have that

1. $\hat{\boldsymbol{\beta}}_n$ and $\hat{\sigma}_n^2$ are *consistent* estimator of $\boldsymbol{\beta}$ and σ^2
2. $\hat{\boldsymbol{\beta}}_n$ is *asymptotically normally* distributed

$$\sqrt{n} (\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \xrightarrow{d} N(0, \sigma^2 \mathbf{V}^{-1}).$$

By approximating $\mathbf{V} \stackrel{a}{\sim} \frac{1}{n} \mathbf{X}'_n \mathbf{X}_n$ we have

$$\hat{\boldsymbol{\beta}}_n \stackrel{a}{\sim} N(\boldsymbol{\beta}, \hat{\sigma}_n^2 (\mathbf{X}'_n \mathbf{X}_n)^{-1}).$$

3.11

Some Questions

- What does mean asymptotic?
- What does happen if the sample size is not so large?

In order to evaluate estimators with 'small' sample sizes, or when the normality assumption on the errors does not hold, we can have recourse to robust techniques like the *bootstrap*, see JWHT.

3.12

3 Goodness of Fit Measures

The Coefficient of determination R^2

The Coefficient of determination is defined as

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\hat{\epsilon}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip})$.

- If $p = 1$ (only one regressor) $R^2 = \rho_{XY}^2$.
- In presence of two nested regression models $\mathcal{M}_h$ and $\mathcal{M}_k$, with $h > k$ – that is of two models characterized by the fact that predictors in $\mathcal{M}_k$ are also predictors in $\mathcal{M}_h$ – we have that $R^2_{\mathcal{M}_h} \geq R^2_{\mathcal{M}_k}$.
- Two *complete* models – that is two models that have the intercept as *free* parameter – can be compared with R^2 only if they have the *same* outcome variable and the *same number* of predictors.

The R^2 is based on the training data set, that is on the same set of data used for model estimation.

Namely, we have

$$\begin{aligned} R^2 &= 1 - \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\frac{1}{n} \sum_{i=1}^n \hat{\epsilon}_i^2}{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2} = \\ &= 1 - \frac{\frac{1}{n} \sum_{i=1}^n \left[y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}) \right]^2}{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2} = \\ &= 1 - \frac{MSE_{training}}{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}. \end{aligned}$$

We can obtain indirect estimates of MSE_{test} , that can also be used for model selection – that is for choosing an ‘optimal’ subset of the p predictors to be included in the model –, by the following mathematical adjustments of $MSE_{training}$

- Mallows’ C_p
- AIC
- BIC
- adjusted R^2

A direct estimate of MSE_{test} can be obtained by means of cross validation, cp. JWHT.

Remark

C_p , AIC , and BIC all have rigorous theoretical justifications that rely on asymptotic theory. Despite its popularity, and even though it is quite intuitive, the adjusted R^2 is not as well motivated in statistical theory as AIC , BIC , and C_p .

3.1 Estimate of MSE_{test} by Mathematical Adjustments of $MSE_{training}$

We define the training Residual Sum of Squares as

$$RSS_{training} = n \cdot MSE_{training}$$

Mallow's C_p

For a linear model fitted with OLS, Mallow's C_p is defined as

$$C_p = \frac{1}{n} (RSS_{training} + 2d\hat{\sigma}^2)$$

where

- $d \leq p$ is the number of predictors (among the p initial predictors) that entered in the linear regression model
- $\hat{\sigma}^2$ is the estimate of σ_ε^2 , variance of the error in the full model containing all p predictors

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon.$$

The penalty $2d\hat{\sigma}^2$ increases with the number d of predictors.

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The Mallow's C_p statistic is sometimes defined as

$$C'_p = \frac{RSS_{training}}{\hat{\sigma}^2} + 2d - n.$$

The two statistics are equivalent with reference to the selection problem; namely, we have

$$C_p = \frac{1}{n} \hat{\sigma}^2 (C'_p + n).$$

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Akaike Information Criterion (AIC)

The *AIC* criterion is used with models estimated with maximum likelihood. In presence of a linear model we have

$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS_{training} + 2d\hat{\sigma}^2)$$

where

- $d \leq p$ is the number of predictors (among the p initial predictors) that entered in the linear regression model
- $\hat{\sigma}^2$ is the estimate of σ_ε^2 , variance of the error component in the full model with all the p predictors.

AIC is proportional to the Mallow's C_p statistic.

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Schwarz Bayesian Information Criterion (BIC)

The *BIC* criterion comes from a Bayesian approach, and is used, like the *AIC*, for models estimated with maximum likelihood. In presence of a linear model it is defined as

$$BIC = \frac{1}{n} (RSS_{training} + \ln(n)d\hat{\sigma}^2)$$

where

- $d \leq p$ is the number of predictors (among the p initial predictors) that entered in the linear regression model
- $\hat{\sigma}^2$ is the estimate of σ_ε^2 , variance of the error component in the full model with all the p predictors.

The penalty $\ln(n)d\hat{\sigma}^2$ term increases with both the sample size as well as the number of predictors.

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Adjusted R^2

$$\text{Adjusted } R^2 = 1 - \frac{RSS_{training}/(n-d-1)}{TSS/(n-1)}$$

where

- d is the number of predictors
- $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ (Total sum of squares) is the total of squared deviations from the sample mean.

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4 Some Checks of the Assumptions of the Linear Model

Satisfaction of assumptions of linear regression models have to be checked before taking any conclusion.

3.23

4.1 Multicollinearity

$\mathbf{X}$ has full column rank $r(\mathbf{X}) = (p + 1)$

Estimation cannot be performed when $\mathbf{X}$ does not have full column rank. In this case statistical softwares first drop one of the collinear variables, that is one of the variables for which an exact linear relationship holds, and then perform estimation.

Therefore, in these situation there are no issues.

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Multicollinearity situations do represent a serious risk.

Multicollinearity

Multicollinearity is present when a set of variables are characterized by a quasi-linear relationship. In this case $r(\mathbf{X}) = (p + 1)$ holds, but inference results are no more reliable.

3.25

100 3. Linear Regression

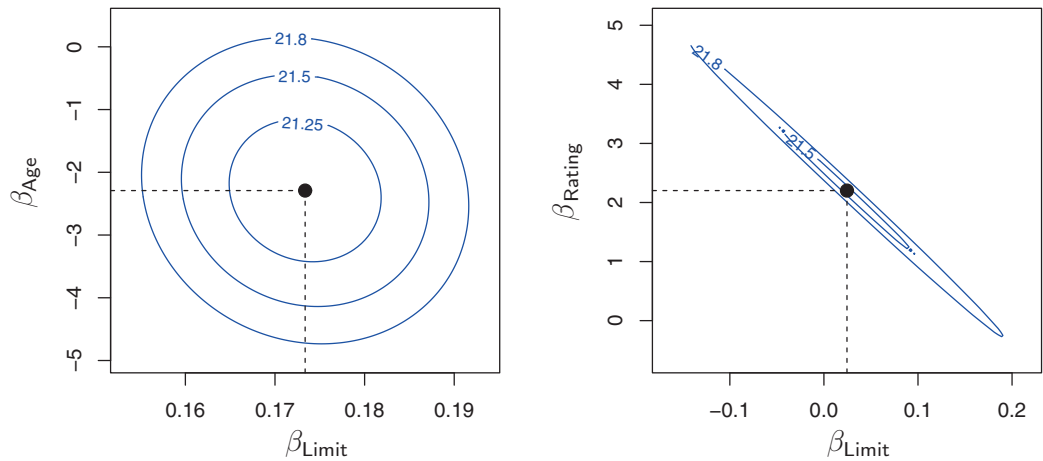


FIGURE 3.15. Contour plots for the RSS values as a function of the parameters β for various regressions involving the **Credit** data set. In each plot, the black dots represent the coefficient values corresponding to the minimum RSS. Left: A contour plot of RSS for the regression of **balance** onto **age** and **limit**. The minimum value is well defined. Right: A contour plot of RSS for the regression of **balance** onto **rating** and **limit**. Because of the collinearity, there are many pairs $(\beta_{\text{Limit}}, \beta_{\text{Rating}})$ with a similar value for RSS.

Figure extracted from James, Witten, Hastie, Tibshirani (2013) An Introduction to Statistical Learning, Springer

3.26

Multicollinearity can be detected by having recourse to the Variance Inflation Factor (*VIF*) statistic

$$VIF(\hat{\beta}_j) = \frac{1}{1 - R_{X_j|X_{-j}}^2}.$$

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With a *VIF* of 1.9 the variance of a coefficient estimate results 90% larger than it would be if that predictor was completely uncorrelated with all the other predictors. *VIF* values larger than 2.5 need attention and values larger than 10 denote problematic situations.

VIF values can be obtained in R with the function `vif` available in the package `car`. Variable selection methods or methods like the lasso, the ridge regression, the principal component or the PLS regression can be adopted, see JWHT.

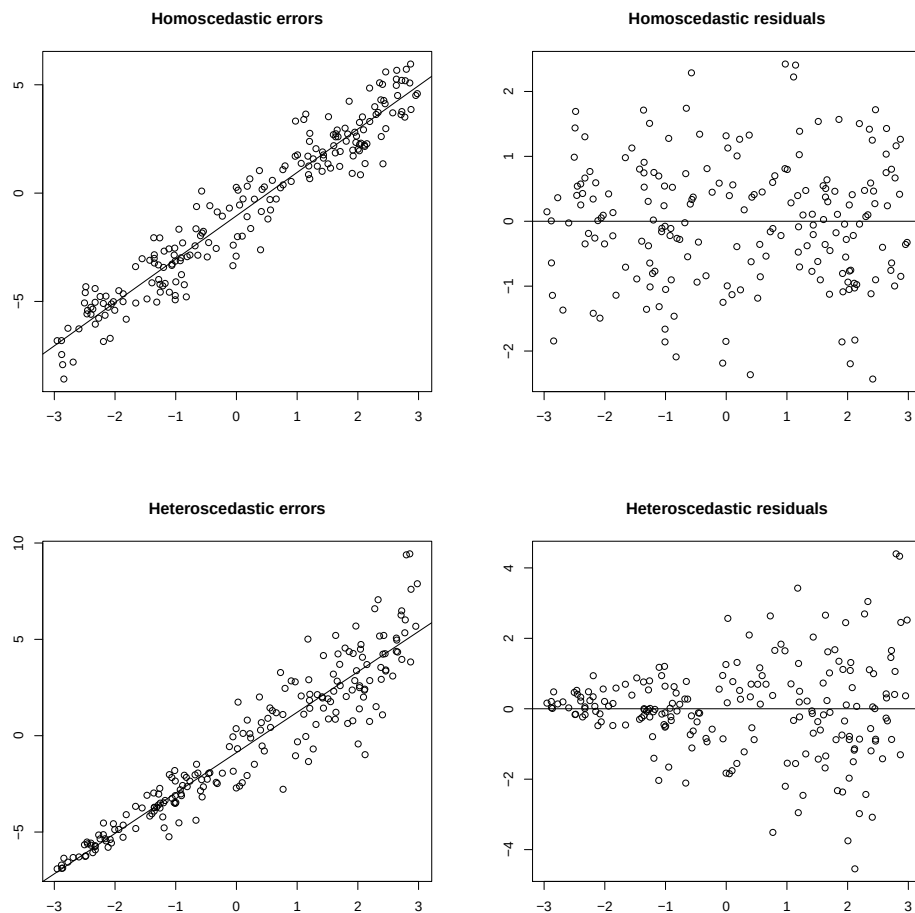
3.28

4.2 Residuals

Graphical inspection of residuals can help the researcher to detect bad situations.

3.29

Assumption: the errors have the same variance for all subjects



3.30

Homoscedastic simulated model

$$y_i = -1 + 2x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(\mu = 0, \sigma^2 = 1)$$

Heteroscedastic simulated model

$$y_i = -1 + 2x_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{independent}}{\sim} N(\mu = 0, \sigma^2 = (0.1 + 0.3 \cdot (x + 3))^2)$$

3.31

Heteroscedastic situations can be detected by having recourse to the Breusch Pagan or the White tests, that are both implemented in the function `bptest` available in the package `lmtest`.

A log transformation of the outcome can be used to remove heteroscedasticity of the multiplicative type, that is present when a model of the type

$$y_i = g(x_i) \exp \eta_i = (a + bx_i) \exp \eta_i,$$

where η_i is an error that has mean zero (conditional upon x_i), is estimated.

Otherwise, generalized least squares can be applied, see e.g. chapter 4 of Verbeek, A Guide to Modern Econometrics, Wiley and Cantaluppi, Computational Laboratory for Economics with R. However, inference on parameters can always be performed by having recourse to robust techniques like the *bootstrap*, see JWHT.

3.32

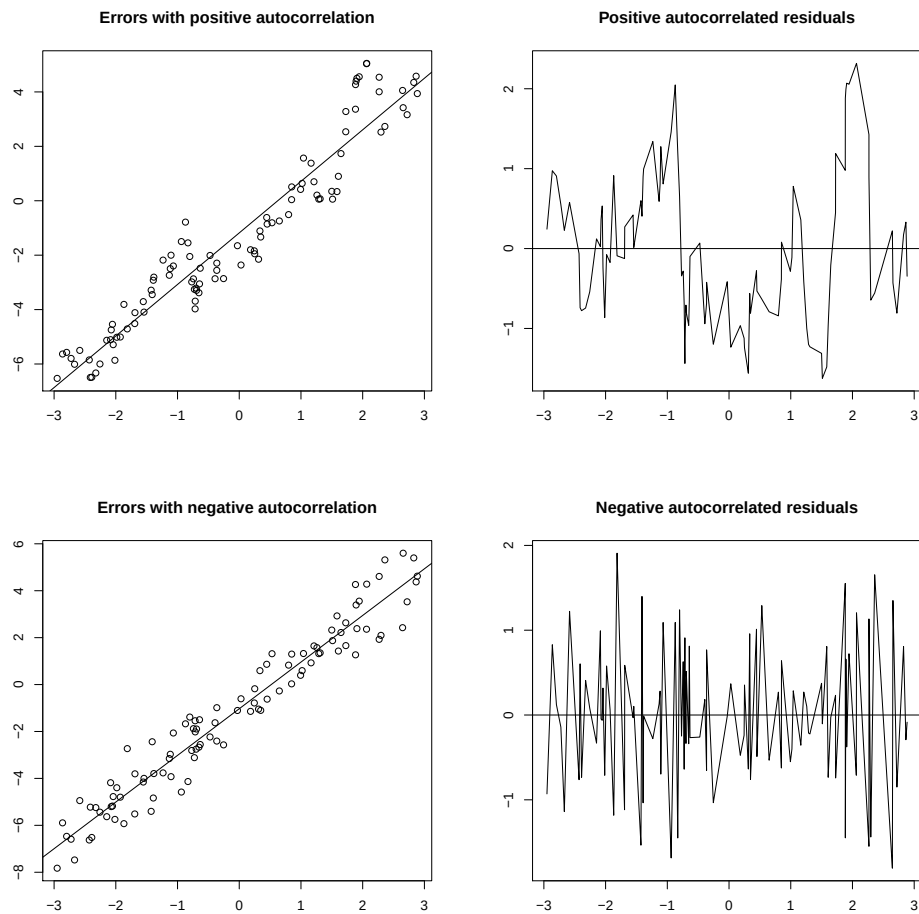
```
# data generation
set.seed(23)
x <- runif(200, -3, 3)
ya <- -1 + 2 * x + rnorm(200)
yb <- -1 + 2 * x + rnorm(200, 0, 0.1 + 0.3 * (x + 3))
# model estimation
outa <- lm(ya ~ x)
outb <- lm(yb ~ x)
layout(matrix(1:4, 2, 2, byrow = TRUE))
plot(ya ~ x, main = "Homoscedastic errors", xlab = "", ylab = "")
abline(outa)
plot(outa$res ~ x, main = "Homoscedastic residuals", xlab = "",
      ylab = "")
abline(lm(outa$res ~ x))
plot(yb ~ x, main = "Heteroscedastic errors", xlab = "", ylab = "")
abline(outb)
plot(outb$res ~ x, main = "Heteroscedastic residuals", xlab = "",
      ylab = "")
abline(lm(outb$res ~ x))
```

3.33

```
library(lmtest)
bptest(outa)
##
## studentized Breusch-Pagan test
##
## data: outa
## BP = 0.28001, df = 1, p-value = 0.5967
bptest(outb)
##
## studentized Breusch-Pagan test
##
## data: outb
## BP = 29.889, df = 1, p-value = 4.575e-08
```

3.34

Assumption: errors are uncorrelated



3.35

Model with positive autocorrelated error

$$y_i = -1 + 2x_i + \varepsilon_i, \quad \varepsilon_i = 0.9 \cdot \varepsilon_{i-1} + u_i, \quad u_i \stackrel{i.i.d.}{\sim} N(\mu = 0, \sigma^2 = 0.25)$$

Model with negative autocorrelated error

$$y_i = -1 + 2x_i + \varepsilon_i, \quad \varepsilon_i = -0.9 \cdot \varepsilon_{i-1} + u_i, \quad u_i \stackrel{i.i.d.}{\sim} N(\mu = 0, \sigma^2 = 0.25)$$

Time series methods can be used to detect and take into account of autocorrelated errors. See e.g. chapters 4 and 8 of Verbeek, A Guide to Modern Econometrics, Wiley and Cantaluppi, Computational Laboratory for Economics with R.

3.36

```
# data generation
set.seed(12)
x <- sort(runif(100, -3, 3))
ea <- arima.sim(list(ar = 0.9), 100, sd = 0.5)
eb <- arima.sim(list(ar = -0.9), 100, sd = 0.5)
ya <- as.numeric(-1 + 2 * x + ea)
yb <- as.numeric(-1 + 2 * x + eb)
# model estimation
```

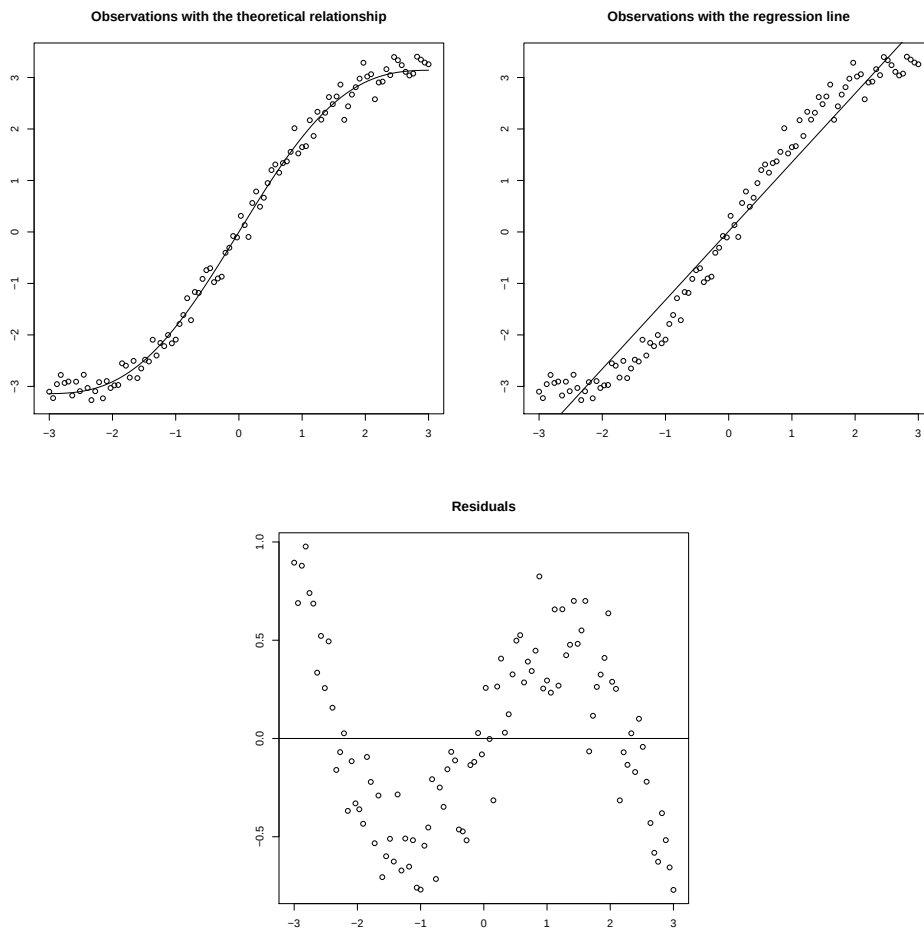
```

outa <- lm(ya ~ x)
outb <- lm(yb ~ x)
layout(matrix(1:4, 2, 2, byrow = TRUE))
plot(ya ~ x, main = "Errors with positive autocorrelation", xlab = "",
      ylab = "")
abline(outa)
plot(outa$res ~ x, type = "l", main = "Positive autocorrelated residuals",
      xlab = "", ylab = "")
abline(lm(outa$res ~ x))
plot(yb ~ x, main = "Errors with negative autocorrelation", xlab = "",
      ylab = "")
abline(outb)
plot(outb$res ~ x, type = "l", main = "Negative autocorrelated residuals",
      xlab = "", ylab = "")
abline(lm(outb$res ~ x))

```

3.37

Autocorrelated residuals can be present when the model is misspecified



3.38

The theoretical relationship

$$y_i = \sin(x_i) + x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(\mu = 0, \sigma^2 = 0.04)$$

was estimated with a linear model.

The problem can be solved by taking into account quadratic or higher order effects or by having recourse to a non-linear specification.

3.39

```
# data generation
set.seed(23)
x <- seq(-3, 3, length = 100)
yteo <- sin(x) + x
y <- yteo + rnorm(100, sd = 0.2)
out <- lm(y ~ x)
layout(matrix(c(1, 1, 2, 2, 3, 4, 4, 5), 2, 4, byrow = TRUE))
plot(y ~ x, main = "Observations with the theoretical relationship",
      xlab = "", ylab = "")
curve(sin(x) + x, add = TRUE)
plot(y ~ x, main = "Observations with the regression line", xlab = "",
      ylab = "")
abline(out)
plot(0, type = "n", xaxt = "n", yaxt = "n", bty = "n", xlab = "",
      ylab = "")
plot(out$res ~ x, main = "Residuals", xlab = "", ylab = "")
abline(lm(out$res ~ x))
```

3.40

Autocorrelated residuals can be present when important variables were excluded from the model

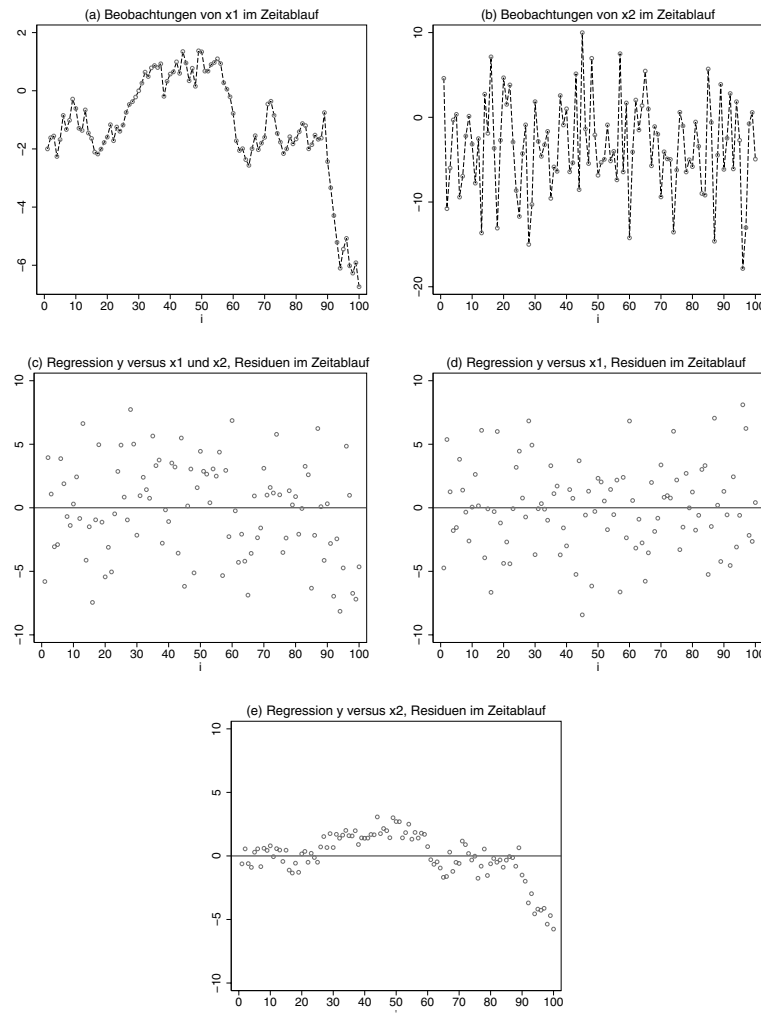


Abb. 3.5. Illustration für autokorrelierte Störungen bei Nichtberücksichtigung von Kovariablen, die einen zeitlichen Trend aufweisen. Die Grafiken a) und b) zeigen die Kovariablen x_1 und x_2 im Zeitablauf. In den Grafiken c)-e) sind die Residuen für die Regressionsmodelle $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ (korrekt spezifiziertes Modell), $y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$ (x_2 nicht berücksichtigt) und $y_i = \beta_0 + \beta_1 x_{i2} + \varepsilon_i$ (x_1 nicht berücksichtigt) zu sehen.

Figure extracted from Fahrmeir L., Kneib T., Lang S. (2007) Regression. Modelle, Methoden und Anwendungen, Springer

The problem can be solved by including also omitted variables.

3.41

3.42

Assumption: errors are normally distributed

Tests for assessing deviations of the distribution of residuals from Normal can be performed.

Graphical inspection of qq-plots and pp-plots give information about deviations on the tails and on the center of the distribution, respectively. See Cantaluppi, Computational Laboratory for Economics with R, EDUCatt.

The R function `fit.cont` available in the package `riskDistributions` performs

a suite of tests in order to assess the distribution type of residuals. See Cantaluppi, Computational Laboratory for Economics with R, EDUCatt.

3.43

Errors are uncorrelated with regressors

Endogeneity issues are present when errors are correlated with regressors.

In Econometrics endogeneity issues are treated by having recourse to instrumental variables estimators, see e.g. Verbeek, A Guide to Modern Econometrics, Wiley.

3.44

4.3 Outliers and Influential Data

Outliers

An outlier is an observation with value very different from other observations (e.g. from the mean value).

For each single variable (predictors or outcome) outliers can be detected by having recourse to boxplots.

Bivariate and multivariate outliers can be detected by having recourse to tests based on the Mahalanobis distance from the centroid.

3.45

Outliers in regression models

Observations that can be considered outliers in a regression model can be detected in the following way

- with reference to the dependent variable we can consider
 - standardized residuals

$$r_i = \frac{\hat{\epsilon}_i}{\hat{\sigma}\sqrt{1-h_{ii}}},$$

where h_{ii} is the so-called leverage, that is the i th element on the diagonal of the hat matrix $\mathbf{H}$.

- studentized residuals

$$r_i^* = \frac{\hat{\epsilon}_{(i)}}{\hat{\sigma}_{(i)}\sqrt{1 + \mathbf{x}_i'(\mathbf{X}_{(i)}'\mathbf{X}_{(i)})^{-1}\mathbf{x}_i}},$$

where $_{(i)}$ refers to the model estimated after having dropped the i th observation;

if $|r_i^*| > 3$ the i th observation is a possible outlier.

- with reference to the whole set of predictors the leverage h_{ii} can be considered; observations with $h_{ii} > \frac{2(p+1)}{n}$ might be problematic.

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Influential Data

Observations, which have simultaneously $|r_i^*| > 3$ and $h_{ii} > \frac{2(p+1)}{n}$ are influential data

An influential observation is, usually, also characterized by a value of the Cook distance

$$D_i = \frac{(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}})'(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}})}{(p+1) \cdot \hat{\sigma}^2}$$

larger than 1.

$_{(i)}$ refers again to the model estimated after having dropped the i th observation;

Studentized residuals, leverages h_{ii} and Cook distances can be obtained in R by applying the functions

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- `rstudent`
- `hatvalues`
- `cooks.distance`

to an `lm` object.

3.48

5 Example. Influential Data Identification in a Bivariate Regression

```
# data generation
set.seed(123)
x <- rnorm(200, 4, 1)
y <- 2 + 0.5 * x + rnorm(200)
# model estimation
out <- lm(y ~ x)
```

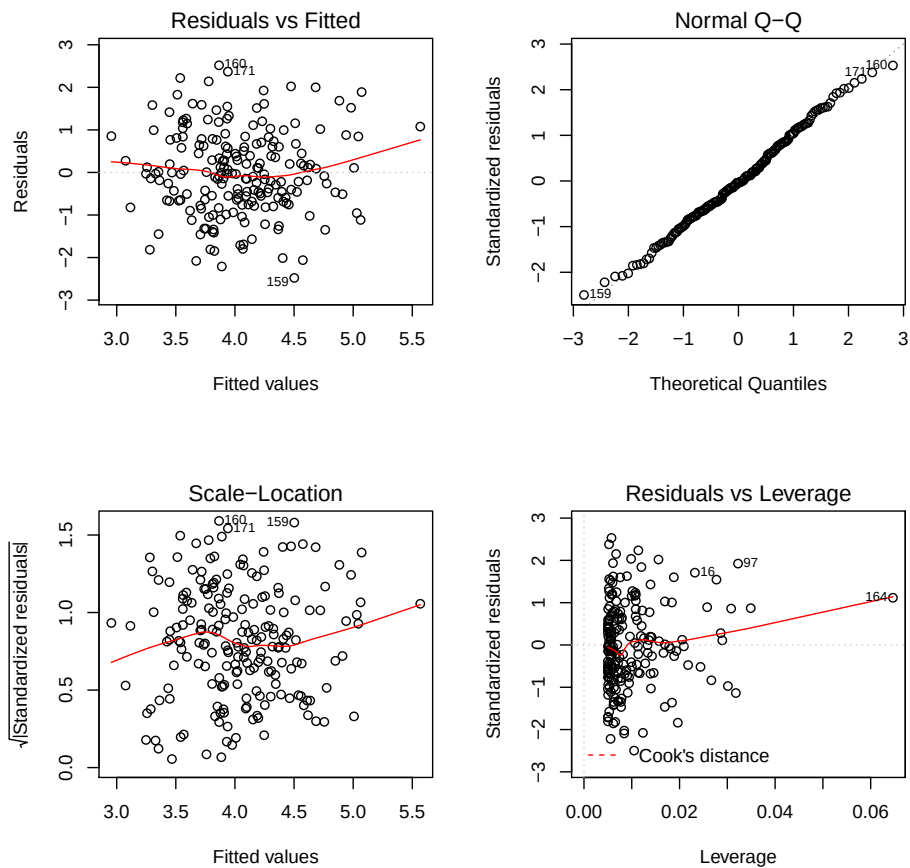
3.49

```
summary(out)
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.47919 -0.65619 -0.03407  0.68037  2.51863
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.15889    0.30762   7.018 3.50e-11 ***
## x             0.47075    0.07501   6.275 2.15e-09 ***
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.998 on 198 degrees of freedom
## Multiple R-squared:  0.1659, Adjusted R-squared:  0.1617
## F-statistic: 39.38 on 1 and 198 DF,  p-value: 2.155e-09
```

3.50

```
layout(matrix(1:4, 2, 2, byrow = TRUE))
plot(out)
```

3.51



3.52

```
# data generation
set.seed(123)
xa <- xb <- xc <- xd <- x <- rnorm(200, 4, 1)
ya <- yb <- yc <- yd <- y <- 2 + 0.5 * x + rnorm(200)
xa <- c(xa, mean(xa))
ya <- c(ya, 10)
xb <- c(xb, 15)
yb <- c(yb, mean(yb))
xc <- c(xc, 15)
yc <- c(yc, 15)
xd <- c(xd, 15)
yd <- c(yd, 9.5)
```

3.53

```
# model estimation
outa <- lm(ya ~ xa)
outb <- lm(yb ~ xb)
outc <- lm(yc ~ xc)
outd <- lm(yd ~ xd)

layout(matrix(1:6, 2, 3, byrow = TRUE))
plot(y ~ x, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4, cex.axis = 2)
abline(out)
plot(ya ~ xa, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4,
```

3.54

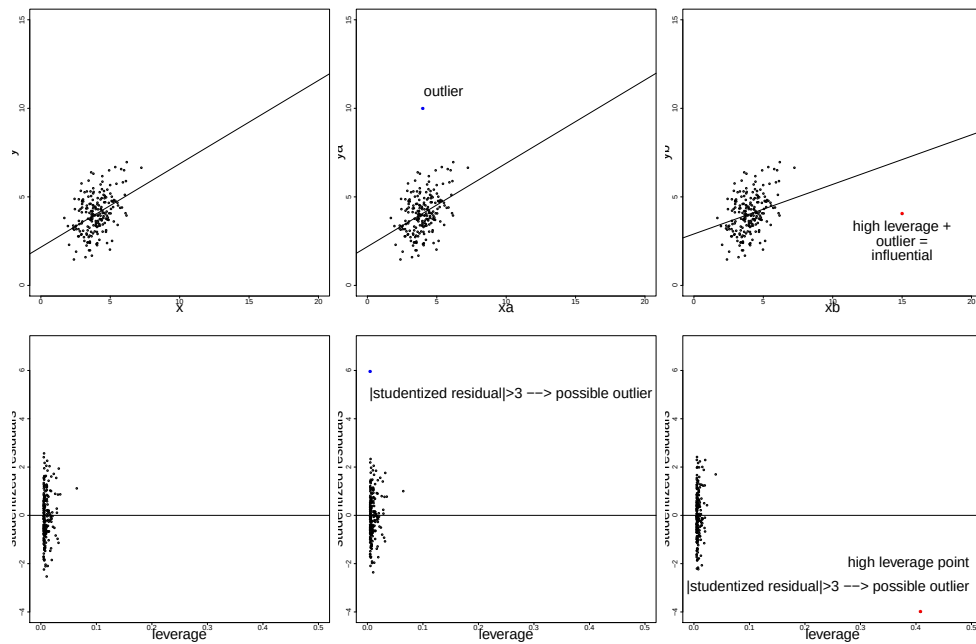


Figure 1: Scatter plots and Studentized residuals versus leverage graphs.

Parameter estimates and R^2 for the three model are:

$\hat{a} = 2.16, \hat{b} = 0.47, \rho^2 = 0.17$ (base model);

$\hat{a} = 2.19, \hat{b} = 0.47, \rho^2 = 0.14$ (model a);

$\hat{a} = 2.9, \hat{b} = 0.28, \rho^2 = 0.1$ (model b)

```

cex.axis = 2)
points(xa[201], ya[201], pch = 16, col = "blue", cex = 2)
abline(outa)
text(5.5, 11, "outlier", cex = 4)
texts <- c("high leverage +", "outlier =", "influential")
plot(yb ~ xb, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4,
     cex.axis = 2)
points(xb[201], yb[201], pch = 16, col = "red", cex = 2)
abline(outb)
sapply(1:3, function(i) text(15, 4 - 0.75 * i, texts[i], cex = 4))

plot(rstudent(out) ~ hatvalues(out), xlim = c(0, 0.5), ylim = c(-4,
7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
plot(rstudent(outa) ~ hatvalues(outa), xlim = c(0, 0.5), ylim = c(-4,
7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
points(hatvalues(outa)[201], rstudent(outa)[201], pch = 16, col = "blue",
      cex = 2)
text(0, 5, "|studentized residual|>3 --> possible outlier", pos = 4,
     cex = 4)
texts <- c("|studentized residual|>3 --> possible outlier", "high leverage point")
plot(rstudent(outb) ~ hatvalues(outb), xlim = c(0, 0.5), ylim = c(-4,
7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
points(hatvalues(outb)[201], rstudent(outb)[201], pch = 16, col = "red",
      cex = 2)
sapply(1:3, function(i) text(0.5, -4 + i, texts[i], pos = 2,
      cex = 4))

```

3.55

3.56

```

layout(matrix(1:6, 2, 3, byrow = TRUE))
plot(y ~ x, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4, cex.axis = 2)
abline(out)
texts <- c("high leverage +", "outlier =", "influential")
plot(yc ~ xc, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4,
      cex.axis = 2)
points(xc[201], yc[201], pch = 16, col = "red", cex = 2)
abline(outc)
sapply(1:3, function(i) text(15, 15 - 0.75 * i, texts[i], cex = 4))
texts <- c("high leverage but", "not outlier =", "not influential")
plot(yd ~ xd, xlim = c(0, 20), ylim = c(0, 15), cex.lab = 4,
      cex.axis = 2)
points(xd[201], yd[201], pch = 16, col = "blue", cex = 2)
abline(outd)
sapply(1:3, function(i) text(15, 13 - 0.75 * i, texts[i], cex = 4))

plot(rstudent(out) ~ hatvalues(out), xlim = c(0, 0.5), ylim = c(-4,
  7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
texts <- c("|studentized residual|>3 --> possible outlier", "high leverage point",
  "influential observation")
plot(rstudent(outc) ~ hatvalues(outc), xlim = c(0, 0.5), ylim = c(-4,
  7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
points(hatvalues(outc)[201], rstudent(outc)[201], pch = 16, col = "red",
  cex = 2)
sapply(1:3, function(i) text(0.5, 4.5 - i, texts[i], pos = 2,
  cex = 4))
texts <- c("|studentized residual|<3 --> no outlier", "high leverage point")
plot(rstudent(outd) ~ hatvalues(outd), xlim = c(0, 0.5), ylim = c(-4,
  7), cex.lab = 4, cex.axis = 2, xlab = "leverage", ylab = "studentized residuals")
abline(h = 0)
points(hatvalues(outd)[201], rstudent(outd)[201], pch = 16, col = "blue",
  cex = 2)
sapply(1:2, function(i) text(0.5, 3 - i, texts[i], pos = 2, cex = 4))

```

3.57

Which cases do have h_{ii} larger than $\frac{2(p+1)}{n}$?

3.58

```

crit <- function(obj) 2 * length(obj$coef) / length(obj$res)
which(hatvalues(out) > crit(out))
##      6      16      18      26      44      70      72      97      108      125      135      139      149      164      174
##      6      16      18      26      44      70      72      97      108      125      135      139      149      164      174
## 196
## 196

which(hatvalues(outa) > crit(outa))
##      6      16      18      26      44      70      72      97      108      125      135      139      149      164      174
##      6      16      18      26      44      70      72      97      108      125      135      139      149      164      174
## 196
## 196

which(hatvalues(outb) > crit(outb))
##      44      72      97      164      201
##      44      72      97      164      201

which(hatvalues(outc) > crit(outc))
##      44      72      97      164      201
##      44      72      97      164      201

which(hatvalues(outd) > crit(outd))
##      44      72      97      164      201
##      44      72      97      164      201

```

3.59

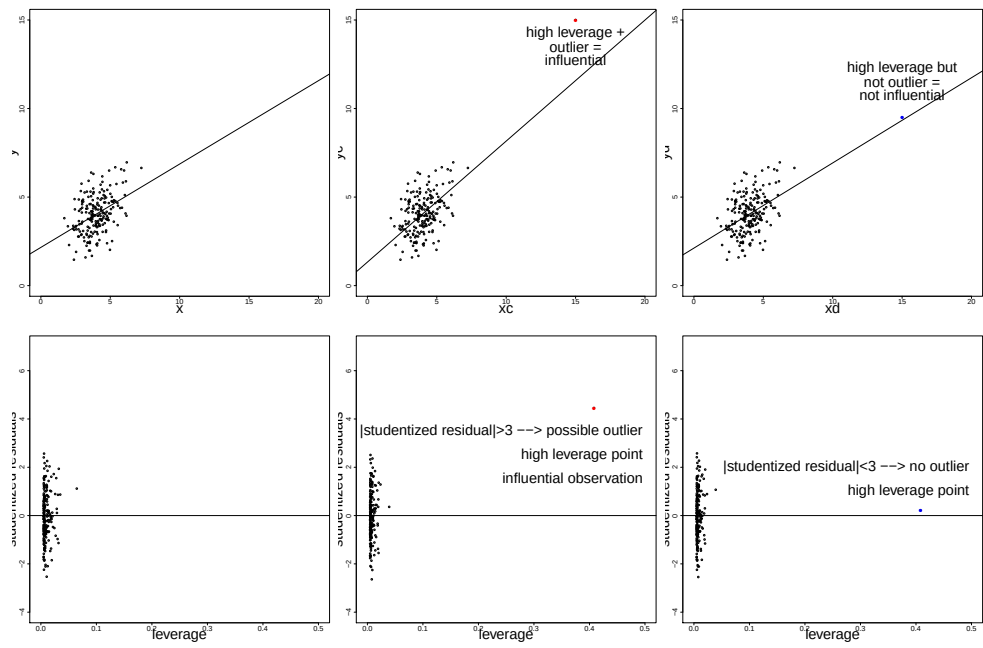


Figure 2: Scatter plots and Studentized residuals versus leverage graphs.

Parameter estimates and R^2 for the three model are:

$\hat{a} = 2.16$, $\hat{b} = 0.47$, $\rho^2 = 0.17$ (base model);

$\hat{a} = 1.33$, $\hat{b} = 0.68$, $\rho^2 = 0.39$ (model c);

$\hat{a} = 2.12$, $\hat{b} = 0.48$, $\rho^2 = 0.26$ (model d)

Which cases do have absolute studentized residuals larger than 3?

```
crit <- function(obj) 2 * length(obj$coef)/length(obj$res)
which(abs(rstudent(out)) > 3)
## named integer(0)
which(abs(rstudent(outa)) > 3)
## 201
## 201
which(abs(rstudent(outb)) > 3)
## 201
## 201
which(abs(rstudent(outc)) > 3)
## 201
## 201
which(abs(rstudent(outd)) > 3)
## named integer(0)
```

3.60

Which cases do have Cook distance larger than 1?

```
crit <- function(obj) 2 * length(obj$coef)/length(obj$res)
which(cooks.distance(out) > 1)
## named integer(0)
which(cooks.distance(outa) > 1)
## named integer(0)
which(cooks.distance(outb) > 1)
## 201
## 201
which(cooks.distance(outc) > 1)
## 201
## 201
which(cooks.distance(outd) > 1)
## named integer(0)
```

3.61

Which cases do have h_{ii} larger than $\frac{2(p+1)}{n}$ and absolute studentized residuals larger than 3 or Cook distance larger than 1?

```
findpossibleinfluential <- function(obj) {
  crit <- 2 * length(obj$coef)/length(obj$res)
  which((abs(rstudent(obj)) > 3) & (hatvalues(obj) > crit) |
        (cooks.distance(obj) > 1))
}
findpossibleinfluential(out)
## named integer(0)
findpossibleinfluential(outa)
## named integer(0)
findpossibleinfluential(outb)
## 201
## 201
findpossibleinfluential(outc)
## 201
## 201
findpossibleinfluential(outd)
## named integer(0)
```

3.62

6 Effect Sizes

Effect sizes can be used as standardized indices that are independent of sample size, in order to quantify the difference between populations or the relationship between explanatory and response variables.

They are also used to perform power analysis.

Reporting effect sizes helps the interpretation of the substantive, as opposed to statistical, significance of a research result.

See e.g. Cohen (1989) Statistical Power Analysis for the Behavioral Sciences, Lawrence Erlbaum Associates, and

<http://stats.idre.ucla.edu/other/mult-pkg/faq/general/effect-size-power/faqhow-is-effect-size-used-in-power-analysis/> and

<http://wiki.biologyscholars.org/@api/deki/files/1587/=Coe2002ItsTheEffectSize.pdf>

3.63

Effect Sizes for F -ratios in regression analysis

For OLS regression the measure of effect size is F which is defined by Cohen as follows.

$$f^2 = \frac{R^2}{1 - R^2} \quad (1)$$

$$f^2 = \frac{\eta^2}{1 - \eta^2}$$

values of f^2 around 0.02, 0.15, and 0.35 are considered as small, medium and large, respectively.

This indication should not be taken as absolute and should be interpreted within the context of the research program.

3.64

Effect Sizes for regression coefficients

In order to interpret coefficients size one can consider:

- standardized coefficients, β_{std} – the coefficients β obtained from estimating a model on the standardized variables.
- partial R^2 – The proportion of residual variation explained by adding the predictor to the constrained model (the full model without the predictor). It is the same as the square of the partial correlation between the predictor and the dependent variable, controlling for all the other predictors in the model

$$R_{partial}^2 = \frac{res.var.constr - res.var.full}{res.var.constr} = \frac{R_{full}^2 - R_{constr}^2}{1 - R_{constr}^2} = \rho_{(x_i | \mathbf{X} \setminus x_i)(y | \mathbf{X} \setminus x_i)}^2$$

where

$$res.var. = \frac{1}{n} \hat{\epsilon}' \hat{\epsilon}.$$

- ΔR^2 – The difference in R^2 between the constrained and full model. It equals the so-called squared semipartial correlation

$$\Delta R^2 = R_{full}^2 - R_{constr}^2 = \rho_{x_i(y | \mathbf{X} \setminus x_i)}^2.$$

As effect size measure Cohen suggests to apply relationship (1) to $R_{partial}^2$.

See e.g. <https://stats.stackexchange.com/questions/64010/importance-of-predictors-in-multiple-regression-partial-r2-vs-standardized?noredirect=1&lq=1>

3.65

The R package `ppcor` can be used to compute partial and semipartial correlations. Semipartial correlations ΔR^2 can be also obtained by applying the function `calc.relimp` available in the package `relaimpo`. It appears in the column `last`, among relative importance metrics. The metric `pmvd` is available, by installing the extended version of this package; `pmvd` can be also used to evaluate relative importance of predictors; see `?calc.relimp` for further info. The `compute.es` R package can be used to compute most commonly encountered effect sizes with their confidence intervals. Not those related to regression coefficients, unfortunately.

3.66